

David Joseph, Ph.D.

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EDUCATION

Ph.D. Max Planck Institute for Multidisciplinary Sciences, Germany 2020–2024

Program: IMPRS-Physics of Biological and Complex Systems (PBCS)

Dissertation: *Optimal Control Methods for Biomolecular NMR Spectroscopy*

Honors: *with summa cum laude*

M.Sc. Indian Institute of Science, Bangalore, India 2018–2020

NMR Research Centre and Solid State and Structural Chemistry Unit (SSCU)

Dissertation: *Phase Modulated Selective NMR Methods for Resolving Overlapping Spin Systems and Rapid Data Acquisition: Application in Metabolomics*

M.Sc. Mahatma Gandhi University, Kerala, India 2013–2018

Institute for Integrated Programmes and Research in Basic Sciences (IIRBS),

M.Sc. Dissertation: *In-silico design and synthesis of novel ATP-citrate lyase inhibitors from naturally occurring 2-hydroxycitric acids*

Honors: CGPA - 9.26/10

B.Sc. Mahatma Gandhi University, Kerala, India 2013–2018

CGPA: 8.4/10

Institute for Integrated Programmes and Research in Basic Sciences (IIRBS)

High School Diploma 2006–2013

Sainik School Kazhakootam, Kerala, India

ACADEMIC APPOINTMENTS

Postdoctoral Associate 2026–2027

Technical University of Munich, Garching, Germany

Postdoctoral Associate 2024–2025

Max Planck Institute for Multidisciplinary Sciences, Göttingen, Germany

Graduate Researcher 2020–2024

Max Planck Institute for Multidisciplinary Sciences, Göttingen, Germany

TEACHING EXPERIENCE

Tutor, NMR für Strukturchemie und Strukturbiologie I Winter Semester 2021/22

Institution: Georg-August-Universität Göttingen

Tutor, NMR für Strukturchemie und Strukturbiologie II

Summer Semester 2022

Institution: Georg-August-Universität Göttingen

Tutor, NMR für Strukturchemie und Strukturbiologie I

Winter Semester 2022/23

Institution: Georg-August-Universität Göttingen

Teaching Assistant, Principles and Applications of NMR Spectroscopy

2020

NPTEL online course, Indian Institute of Science, Bangalore, India

MENTORING EXPERIENCE

Supervision of Master's Students

Supervised master's students Theresa H. and Marvin L. at MPI-NAT, Göttingen, and Anju T.S. and Sreenath S.V. at IIRBS for M.Sc. projects.

PUBLICATIONS

Peer-Reviewed Research Articles

17. **Joseph, D.**, Kümmerle, R., Freytag, N., Griesinger, C. "Low-power optimal control pulse sequences enhance the sensitivity of 5 mm triple resonance cryogenic probe optimized for proton-detection at 1.2 GHz NMR instruments." *J. Magn. Reson.* 2026, 389, 108081.
16. **Joseph, D.**, Tekwani Movellan, K., Becker, S., Andreas, L. B., Griesinger, C. "Optimal control pulses for Bloch-Siegert shift free NMR experiments." *J. Magn. Reson.* 2026, 386, 108030.
15. M. Schiavina, **Joseph, D.**, Griesinger, C., Felli, I. C., Pierattelli, R. "¹⁵N Optimal Control Pulses: An Efficient Approach to Boost ¹³C-Detected NMR Experiments." *J. Magn. Reson.* 2025, 381, 107972.
14. **Joseph, D.**, Griesinger, C. "Optimal control pulses for the 1.2-GHz (28.2-T) NMR spectrometers." *Sci. Adv.* 2023, 9, ead1133.
13. Jijo, A., Cheredath, A., Uppangala, S., Lakshmi, R. V., **Joseph, D.**, et al. "ICSI in non-male factor infertility patients does not alter metabolomic signature in sibling embryos as evidenced by sensitivity enhanced nuclear magnetic resonance (NMR) spectroscopy." *PLoS One* 2022, 17, e0273321.
12. Patel, A. K.[†], **Joseph, D.**[†], Atreya, H. S., Suryaprakash, N. "A One-Dimensional Phase-Modulated (PM) NMR Experiment for Differentiating Spin Systems in Small Molecules Mixtures: With Application to Chiral Discrimination and Bicomponent Organic Systems." *Anal. Chem.* 2022, 94, 16555–16559.
11. Cheredath, A., Uppangala, S., C. S, A., et al. "Combining Machine Learning with Metabolomic and Embryologic Data Improves Embryo Implantation Prediction." *Reprod. Sci.* 2022, 30, 984–994.
10. Sasikumar, P. G., Sudarshan, N. S., Adurthi, S., et al. "PD-1 derived CA-170 is an oral immune checkpoint inhibitor that exhibits preclinical anti-tumor efficacy." *Commun. Biol.* 2021, 4, 699.

9. Raza, T., Dhaka, N., **Joseph, D.**, Dadhwal, P., Kakita, V. M. R., Atreya, H. S., Mukherjee, S. P. “Insights into the NF- κ B-DNA Interaction through NMR Spectroscopy.” *ACS Omega* 2021, 6, 12877–12886.
8. **Joseph, D.**, Sukumaran, S., Chandra, K., et al. “Rapid nuclear magnetic resonance data acquisition with improved resolution and sensitivity for high-throughput metabolomic analysis.” *Magn. Reson. Chem.* 2021, 59, 300–314.
7. Asampille, G., Cheredath, A., **Joseph, D.**, Adiga, S. K., Atreya, H. S. “The utility of nuclear magnetic resonance spectroscopy in assisted reproduction.” *Open Biol.* 2020, 10, 200092.
6. Abhijith, B. L., Mohan, M., **Joseph, D.**, Haleema, S., Aboul-Enein, H. Y., Ibneusaud, I. “Capillary zone electrophoresis for the analysis of naturally occurring 2-hydroxycitric acids and their lactones.” *J. Sep. Sci.* 2017, 40, 3351–3357.

Submitted Research Articles

5. Dhaka, N., Sareen, J., Raza, T., Chauhan, N., **Joseph, D.**, Mukherjee, S. “NF- κ B dimer exchange on the promoter facilitates transcription activation of NF- κ B repressed IRF3 target genes – a plausible mechanism.” *Submitted to Nucleic Acids Research*.

Articles in pipeline

4. **Joseph, D.**, “Physics-Informed Neural Network for Conformational State Prediction from NMR Spectra.” 2026.
3. **Joseph, D.**, “Pulsar Studio: A Graphical Interface for Quantum Optimal-Control Pulse Design.” 2026.
2. **Joseph, D.**, “Pulsar: Pulse Design Library for Spin Control Algorithms and Rollout.” *Package is available at <https://github.com/DaJo2025/Pulsar.jl>* 2026.

Other Scholarly Contributions

1. Arthanari, H., Chary, K. V. R., Szyperski, T., et al. “Hanudatta S. Atreya (1974–2020).” *Magn. Reson. Chem.* 2021, 59, 201–212.

PRESENTATIONS

Invited Talks

1. FGNMR 2024 (Rostock, Germany). “Optimal control pulses for the 1.2-GHz (28.2-T) NMR spectrometers.” Ernst Award lecture.
2. NAN Seminar 2023 (UW Madison, via Zoom). “Low power optimal control pulses improve the performance of multidimensional biomolecular solution NMR experiments at ultrahigh-field (1.2 GHz) spectrometers.”

Oral Presentations

IMPRS Infinity 2024 (Göttingen, Germany); EUROMAR 2024 (Bilbao, Spain); ENC 2024 (Asilomar, USA); EMBO Course 2023 (Berhampur, India); GRS 2023 (Vermont, USA); ENC 2023 (Asilomar, USA); IMPRS Infinity 2022 (Göttingen, Germany); FGNMR 2022 (Karlsruhe,

Germany); 26th NMRS 2020 (Rajkot, India).

Poster Presentations

ICMRBS 2024 (Seoul, South Korea); EUROMAR 2022 (Utrecht, Netherlands); 12th International Conference “NMR: a tool for biology” (Paris, France).

HONORS & AWARDS

2024 Ernst Award, Division of Magnetic Resonance, German Chemical Society for best paper
2024 JMR/JMRO Young Scientist Award, EUROMAR
2024 Global NMR Online Conference Award
2024 ICMRBS Student Travel Award
2024 ENC Student Travel Stipend
2022 EUROMAR Travel Grant
2018 Qualified Graduate Aptitude Test in Engineering (GATE), Chemical Sciences
2016 Qualified CSIR National Eligibility Test (NET) for Junior Research Fellowship (JRF) and Lectureship, Chemical Sciences (all India rank of 66)

ACADEMIC SERVICE

2023–2025 Distribution of NMR magnets to lab members, MPI-NAT, Göttingen
2022–2025 Maintenance of 1.2 GHz spectrometer, MPI-NAT, Göttingen
2021–2023 Maintenance of 600 and 950 MHz spectrometers, MPI-NAT, Göttingen
2018–2020 Maintenance of 800 MHz NMR spectrometer, IISc Bangalore
2023–2025 Co-organizer, AMPERE Café, MPI-NAT, Göttingen
2025– Reviewer for Science advances